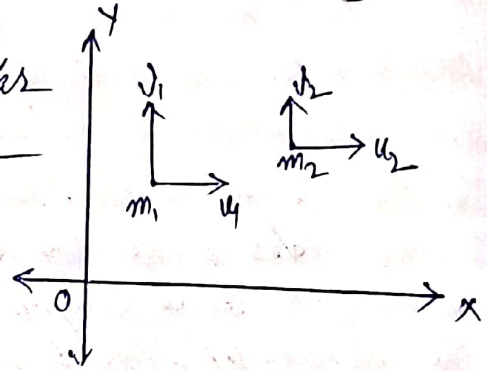


3) Problem 2:- Prove that the kinetic energy of two particles of masses m_1 & m_2 moving in a plane is

$\frac{1}{2} (m_1 + m_2) u^2 + \frac{1}{2} \frac{m_1 m_2}{m_1 + m_2} v^2$, where u is the velocity at the centre of mass of the particle and v is the velocity of either of them relative to the other. [20, 33]

Ans. Referred to a set of rectangular axes in the plane of motion of the masses, let u_1, v_1 & u_2, v_2 be the components of the masses m_1 & m_2 respectively parallel to the axes of X & Y .



Let us denote the components of the velocity v of the centre of masses by U_1 and V_1 . Then derive

$$(m_1 + m_2) U_1 = m_1 u_1 + m_2 u_2 \quad \text{--- (i)}$$

$$(m_1 + m_2) V_1 = m_1 v_1 + m_2 v_2 \quad \text{--- (ii)}$$

$$\text{Also, } U_1 + V_1 = v \quad \text{--- (iii)}$$

$$\times (u_1 - u_2) + (v_1 - v_2) = v \quad \text{--- (iv)}$$

v being the velocity of either mass relative to the other.

Squaring (i) & (ii) and adding.

$$(m_1 u_1 + m_2 u_2)^2 + (m_1 v_1 + m_2 v_2)^2 = (m_1 + m_2)^2 (U_1^2 + V_1^2) = (m_1 + m_2)^2 v^2 \quad \text{--- (v)}$$

Again, multiplying (iv) by $m_1 m_2$ and adding this to (v), we have

$$m_1 m_2 (u_1 - u_2)^2 + m_1 m_2 (v_1 - v_2)^2 + (m_1 u_1 + m_2 u_2)^2 + (m_1 v_1 + m_2 v_2)^2 = m_1 m_2 v^2 + (m_1 + m_2)^2 v^2$$

$$m_1 m_2 (u_1^2 + u_2^2) + m_1 m_2 (v_1^2 + v_2^2) - 2 m_1 m_2 (u_1 u_2 + v_1 v_2) + m_1^2 (u_1^2 + v_1^2) + m_2^2 (u_2^2 + v_2^2) + 2 m_1 m_2 (u_1 u_2 + v_1 v_2) = m_1 m_2 v^2 + (m_1 + m_2)^2 v^2$$

$$\text{or, } m_1 (u_1^2 + v_1^2) / (m_1 + m_2) + m_2 (u_2^2 + v_2^2) / (m_1 + m_2) = m_1 m_2 v^2 / (m_1 + m_2) + (m_1 + m_2) v^2$$

$$\text{or } (m_1 + m_2) \left\{ 2 \times \text{K.E. of } m_1 + 2 \times \text{K.E. of } m_2 \right\} = m_1 m_2 v^2 + (m_1 + m_2)^2 v^2$$

$$\text{or, } \text{K.E. of } m_1 + \text{K.E. of } m_2 = \frac{1}{2} (m_1 + m_2) v^2 + \frac{m_1 m_2 v^2}{m_1 + m_2}$$

Problem 9:- A uniform chain of length $2a$ is hung over a smooth peg so that length of it on two sides are $(a+b)$ and $(a-b)$. Motion is allowed to ensue. Show that the chain leaves the peg at the end of time $\sqrt{\frac{a}{g}} \log \left\{ \frac{a+\sqrt{a^2-b^2}}{b} \right\}$. (Q2)

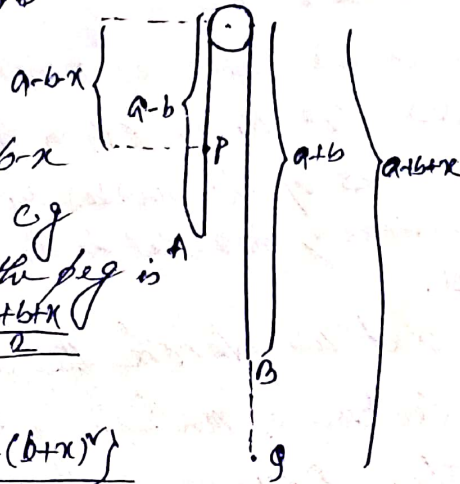
Ans. Let m be mass per unit length of the chain and x be its displacement at any time t .

Then the length of the chain on the two sides of the peg are $a-b-x$ and $a+b+x$ and depth of the c.g. of the string at time t below the peg is

$$\bar{x} = \frac{(a-b-x)m \cdot \frac{a-b-x}{2} + (a+b+x)m \cdot \frac{a+b+x}{2}}{2km}$$

$$= \frac{(a-b+x)^2 + (a+b+x)^2}{4a} = \frac{2\{a^2 + (b+x)^2\}}{4a}$$

$$= \frac{a^2 + (b+x)^2}{2a} \quad \text{--- (i)}$$



Initially c.g. of the string below the peg, we obtain by putting $x=0$ in (i), we have

$$\bar{x}_0 = \frac{a^2 + b^2}{2a} \quad \text{--- (ii)}$$

$$\text{Hence Work done} = 2km g (\bar{x} - \bar{x}_0)$$

$$= 2km g \left\{ \frac{a^2 + (b+x)^2}{2a} - \frac{a^2 + b^2}{2a} \right\} \cdot \frac{1}{2a}$$

$$= mg \{x^2 + 2bx\}$$

Again change in K.E. of the chain

$$= \frac{1}{2} \cdot 2km \dot{x}^2 - 0 = am \dot{x}^2$$

By the principle of energy, we have

Change in K.E. = Work done

$$\therefore am \dot{x}^2 = mg \{x^2 + 2bx\}$$

$$\therefore \frac{dx}{dt} = \sqrt{\frac{g}{a}} \sqrt{(x+b)^2 - b^2}$$

$$\therefore \int_0^{a-b} \frac{dx}{\sqrt{(x+b)^2 - b^2}} = \sqrt{\frac{g}{a}} \int_0^T dt$$

[Let T be the time taken by the particle to reach the end of the chain]

$$\text{on } \sqrt{g/a} T = \left[\ln \left(\frac{a+b+\sqrt{(a+b)^2-b^2}}{b} \right) \right]_0^{a-b}$$

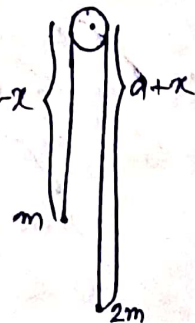
$$a, T = \sqrt{a/g} \left[\ln \left\{ \frac{a+\sqrt{a^2-b^2}}{b} \right\} \right], \text{ or } \boxed{T = \sqrt{a/g} \ln \left\{ \frac{a+\sqrt{a^2-b^2}}{b} \right\}}$$

Problem 4. Two particles of masses m and $2m$ are attached to the two ends of a uniform chain of mass m and length $2a$. The chain is hung over a smooth peg so that its lengths on the two sides are equal and is then allowed to fall. Show that the chain leaves the peg after a time $\sqrt{4a/g} \ln(2+\sqrt{3})$.

Ans. The mass per unit length of the chain is $m/2a$.

Initially, the chain was over the smooth peg so that its length on the either sides are equal. Let x be its displacement in time t . $a-x$

Then the portions of the chain on the two sides of the peg are $a-x$ & $a+x$.



Now, the depth of the c.g. of the system below the peg is

$$\begin{aligned} \bar{x} &= \frac{(a-x) \cdot \frac{m}{2a} \cdot \frac{a-x}{2} + (a+x) \cdot \frac{m}{2a} \cdot \frac{a+x}{2} + m(a-x) + 2m(a+x)}{4m} \\ &= \frac{(a-x)^2 + (a+x)^2 + 4 \cdot 3a^2 + 4ax}{4 \cdot 4a} \\ &= \frac{2(a^2+x^2) + 12a^2 + 4ax}{4 \cdot 4a} = \frac{7a^2 + 2ax + x^2}{8a} \end{aligned}$$

The initial depth of the system below the fixed peg

$$\bar{x}_0 = 7a/8$$

$$\text{Now the work done} = (m+m+2m) \cdot g \cdot \frac{2ax+x^2}{8a}$$

Also change in K.E of the chain = $\frac{1}{2} \cdot 4m \cdot \dot{x}^2 = 2m \dot{x}^2$
By the principle of energy.

$$2m \dot{x}^2 = 4mg \frac{2ax+x^2}{8a}$$

$$\text{or, } \dot{x}^2 = \frac{g}{a} \cdot \frac{2ax+x^2}{4}$$

$$\text{or, } \dot{x} = \sqrt{\frac{g}{4a}} \sqrt{(a+x)^2 - a^2}$$

$$\text{or } \int_0^a \sqrt{\frac{g}{4a}} dt = \int_0^a \frac{dx}{\sqrt{(a+x)^2 - a^2}}$$

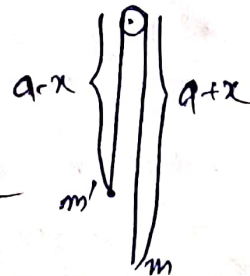
$$\text{or, } \sqrt{\frac{g}{4a}} T = \left[\ln \left(\frac{a+x+\sqrt{(a+x)^2 - a^2}}{a} \right) \right]_0^a = \ln(2+\sqrt{3})$$

$$\text{or, } \sqrt{T} = \sqrt{4a/g} \ln(2+\sqrt{3})$$

§ Problem 5. A uniform string of mass M and length $2a$ is placed symmetrically over a smooth peg and has particles of masses m and m' attached to its ends. Show that, when the string runs off the peg, its velocity is $\left\{ \frac{(M+2(m-m'))g}{M+m+m'} \right\}^{1/2}$ $m > m'$.

Ans. The mass per unit length of the string is $M/2a$.

Initially, the string is pulled symmetrically over the peg, let x be the displacement in time t . Then the length of the string on the two sides of the peg are $a-x$ and $a+x$.



Now the depth of the C.G. of the system below the fixed peg

$$= \bar{x} = \frac{(a-x)m' + (a+x)m + (a-x) \cdot \frac{M}{2a} \cdot \frac{(a-x)}{2} + (a+x) \cdot \frac{M}{2a} \cdot \frac{(a+x)}{2}}{m' + m + M}$$

Then the initial depth of the C.G. of the system below the fixed peg $= \bar{x}_0 = \frac{am' + am + \frac{M}{2a} \cdot a^2}{m' + m + M}$

$$\text{Now work done} = (m' + m + M)g(\bar{x} - \bar{x}_0) \\ = (m' + m + M)g \cdot \frac{(m-m')x + \frac{M}{2a}x^2}{(m' + m + M)}$$

$$= \frac{g}{2a} \{ 2a(m-m')x + Mx^2 \}$$

Now change in KE of the chain $= \frac{1}{2}(m' + m + M)v^2$ by the principle of energy,

$$\frac{1}{2}(m' + m + M)v^2 = \frac{g}{2a} \{ 2a(m-m')x + Mx^2 \}$$

$$\therefore v = \sqrt{\frac{g}{2a} \left\{ \frac{2a(m-m')x + Mx^2}{m' + m + M} \right\}}$$

which gives the velocity of the string at x distance to placement.

So, when the string runs off the peg, its displacement will be $x=a$ and corresponding velocity is

$$v_{x=a} = \sqrt{\frac{g}{2a} \cdot \frac{2a(m-m') + Ma^2}{m' + m + M}} = \left\{ \frac{M+2(m-m')}{M+m+m'} \cdot g \right\}^{1/2}$$

3 Problem:- An engine works at a constant power H units of work per second. It pulls a train of total mass M against a constant resistance R . Show that the train acquires Velocity V in time $\left(\frac{MH}{RV} \log \frac{H}{H-RV} - \frac{MV}{R} \right)$ seconds. 20

Ans. At any instant t , let P be the force exerted by the engine and v be the velocity of the train.

Then rate of doing work is $Pv = H$ (given)
 i.e. $P = \frac{H}{v}$.

The resistance being R , the resultant force acting on the engine is $P - R = \frac{H}{v} - R$

Thus equation of motion is

$$M \frac{dv}{dt} = \frac{H}{v} - R = \frac{H - Rv}{v}$$

$$\text{or } dt = \frac{Mv dv}{H - Rv}$$

If T is the time required to acquire a velocity V

$$\int_0^T dt = \int_0^V \frac{Mv dv}{H - Rv}$$

$$\text{or } T = \frac{M}{R} \int_0^V \frac{H - (H - Rv)}{H - Rv} dv$$

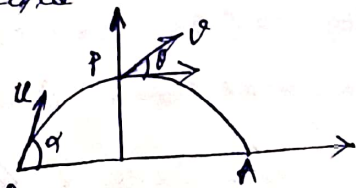
$$= \frac{M}{R} \int_0^V \left[\frac{H}{H - Rv} - 1 \right] dv$$

$$= \frac{M}{R} \left[\frac{H}{R} \log \frac{1}{H - Rv} - v \right]_0^V$$

$$\boxed{T = \frac{MH}{RV} \log \frac{H}{H - RV} - \frac{MV}{R}}$$

Problem 7:- For a particle of mass m projected in vacuum from the horizontal ground with a velocity u at an angle α to the horizon, ~~show that~~ the sum of K.E & P.E of the particle at any position of its path is constant.

Ans. Let a particle of mass m be projected from a point O on the ground with a velocity u making an angle α to the horizon. At any time t , let P be the position of the particle at a height h above the ground and let v be the velocity of the particle at P making an angle θ with the horizontal direction. The force acting on the particle during the entire motion is the weight mg of the particle acting vertically downwards. So the acceleration of the particle is g in the vertically downwards direction and it has no component in the horizontal direction.



Thus, the horizontal component of velocity of the particle remains unchanged during the motion and we've

$$v \cos \theta = u \cos \alpha \quad \text{--- (i)}$$

Considering the vertical motion, we've at P ,

$$v \sin \theta = u \sin \alpha - g h \quad \text{--- (ii)}$$

Squaring (i) and adding with (ii), we've

$$v^2 \cos^2 \theta + v^2 \sin^2 \theta = u^2 (\cos^2 \alpha + \sin^2 \alpha) - 2gh$$

$$\text{or, } v^2 = u^2 - 2gh \quad \text{--- (iii)}$$

$$\text{K.E of the particle at } P = \frac{1}{2} m v^2 = \frac{1}{2} m (u^2 - 2gh)$$

$$\text{and P.E of the particle at } P = mgh$$

$$\text{Then K.E + P.E} = \frac{1}{2} m u^2 - mgh + mgh = \frac{1}{2} m u^2$$

which is independent of h

When the particle is at O , its K.E = $\frac{1}{2} m u^2$

$$\text{P.E} = mgy_0 = 0$$

$$\therefore \text{So, K.E + P.E} = \frac{1}{2} m u^2$$

So, we find that the sum of the K.E & P.E is constant at any position of the particle.

Problem 8:- A body of mass M is propelled in a st. line by an engine producing energy at a constant rate P against a resistance κv^3 , where v is the velocity & κ is a constant. Prove that the space s described from rest is given by

$$\frac{3\kappa s}{M} = -\log\left(1 - \frac{\kappa v^3}{P}\right)$$

Ans. Power P is given by $P = F \cdot v$ i.e. $F = \frac{P}{v}$
Equation of motion is

$$M \cdot v \frac{dv}{ds} = F - \kappa v^3 = \frac{P}{v} - \kappa v^3 = \frac{P - \kappa v^3}{v}$$

$$\text{or } \int_0^s \frac{ds}{M} = \int_0^v \frac{3\kappa v^2 dv}{P - \kappa v^3}$$

$$\text{or } \frac{3\kappa s}{M} = - \left[\log(P - \kappa v^3) \right]_0^v$$

$$\boxed{\frac{3\kappa s}{M} = -\log\left(1 - \frac{\kappa v^3}{P}\right)}$$